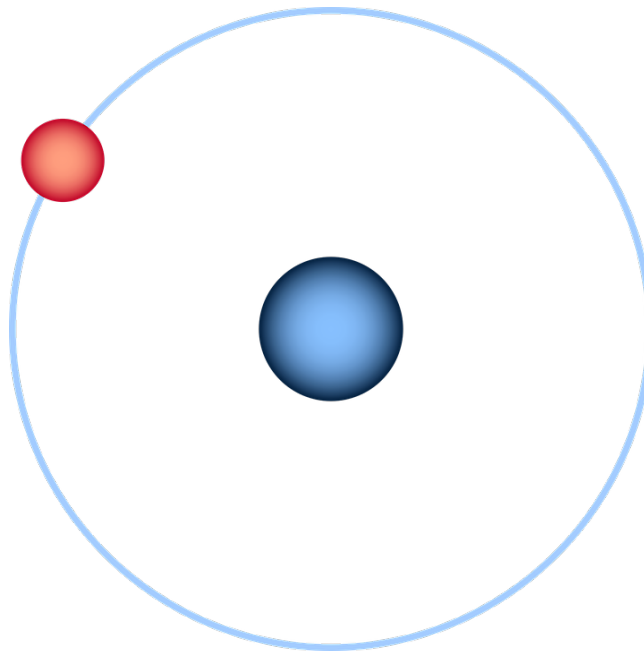


# Electrical System Design for a Hydrogen Solo Circumnavigation Aircraft

Bradley Blaxall  
CID: 02061041  
Systems and Propulsion

---



**Academic Supervisor:** Dr Sebastian Eastham  
**Department:** Department of Aeronautics  
**Course:** MEng in Aeronautics (H401)  
**Module:** AERO 60004  
**Academic Year:** 2023/2024  
**Date:** 17/06/2024

Department of Aeronautics  
South Kensington Campus  
Imperial College London  
London SW7 2AZ  
U.K.

## **Abstract**

This report outlines the design of an electrical system for a hydrogen fuel cell aircraft with a mission of a solo circumnavigation with a solo pilot. The process of designing the electrical system differs greatly from traditional aircraft due to the electrical propulsive system used. The process begins with an estimation of total power needed from each sub system, with the total maximum continuous power requirement being 417 kW. The systems electrical components: cables, DC/DC step down converters and buses, are modelled to allow for weight estimates. Cables were found to be the largest contributor to mass with 64 % of total system mass. The influence of redundancy and operation was explored, resulting in the duplication of most components to allow safe flight in the event of a failure but increasing the mass of the system. Finally key design parameters were chosen such as the fuel cell voltage of 10 kV since this resulted in low mass, fuel cell redundancy and not exceeding the thermal constraints set by insulator material. The electrical components were placed on the aircraft and their positions optimised around the propulsion electrical components as they had the largest impact on overall mass. The final electrical system mass was 423 kg exceeding the initial estimate for this design iteration. The reason for the initial underestimate coming from a lack of knowledge of the location of sub systems at the design stage which impacts heavily the overall mass of the electrical system. This report shows there is a strong viability for an electrical system design for this aircraft and the mission.

**Table 1.** All functional area technical teams with team members listed.

| <b>Subteam</b>                            | <b>Member</b>                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Project coordination                      | Gonzalo Montenegro<br>Ben Tsow<br>Etido Enebeli                                                                     |
| Aerodynamics                              | Ray Borui Ji<br>Tianyu Zou<br>Mark Nedaskovski<br>Layton Richardson                                                 |
| Structural design, manufacturing and CAD  | Devraj Katkoria<br>Shreya Seshan<br>Jason Yuxiong Zhao<br>Michael Tolley                                            |
| <b>Systems and propulsion</b>             | Sita Mandakini Nair<br>Noah Collins<br>Hasti Jalal-Aldeen<br>Dominic Coyde<br><b>Bradley Blaxall</b><br>Mohamed Ali |
| Flight mechanics, performance and control | Andrei Leon<br>Adam Meziane<br>Richard Buscaglia<br>Michelle Tan<br>Jude Mbroh                                      |
| Flight simulation and testing             | Tom Gordon<br>Daniel Malekyazdi<br>Keanu Wamae                                                                      |

# Contents

|                                                          |            |
|----------------------------------------------------------|------------|
| <b>Table of Contents</b>                                 | <b>iii</b> |
| <b>List of Figures</b>                                   | <b>iv</b>  |
| <b>List of Tables</b>                                    | <b>v</b>   |
| <b>1 Introduction</b>                                    | <b>1</b>   |
| <b>2 Power Sizing</b>                                    | <b>1</b>   |
| 2.1 Propulsion . . . . .                                 | 1          |
| 2.2 Avionics . . . . .                                   | 1          |
| 2.3 Environmental Control Systems . . . . .              | 2          |
| 2.4 Thermal Management . . . . .                         | 2          |
| 2.5 Fuel Cell Compressor . . . . .                       | 3          |
| 2.6 Summary . . . . .                                    | 3          |
| <b>3 Electrical Component Sizing</b>                     | <b>3</b>   |
| 3.1 Cables . . . . .                                     | 4          |
| 3.2 Busbars . . . . .                                    | 5          |
| 3.3 Voltage Converters . . . . .                         | 5          |
| <b>4 Redundancy and Operation</b>                        | <b>5</b>   |
| 4.1 Redundancy . . . . .                                 | 6          |
| 4.2 Operation . . . . .                                  | 7          |
| <b>5 Electrical System Design</b>                        | <b>7</b>   |
| 5.1 Component Selection . . . . .                        | 8          |
| 5.2 Fuel Cell Voltage . . . . .                          | 8          |
| 5.3 Electrical Component Position Optimisation . . . . . | 10         |
| 5.4 Summary . . . . .                                    | 11         |
| <b>6 Conclusion</b>                                      | <b>12</b>  |
| <b>References</b>                                        | <b>13</b>  |
| <b>A Cable Insulator Material Properties</b>             | <b>15</b>  |

**List of Figures**

|   |                                                                                      |    |
|---|--------------------------------------------------------------------------------------|----|
| 1 | Layout of avionics systems on the aircraft . . . . .                                 | 2  |
| 2 | Electrical systems layout with redundancy considerations . . . . .                   | 6  |
| 3 | Systems layout on the aircraft . . . . .                                             | 9  |
| 4 | Electrical system mass as fuel cell voltage varies . . . . .                         | 9  |
| 5 | Optimised electrical component locations . . . . .                                   | 10 |
| 6 | 3D plot showing system mass with varying main bus and motor power converter position | 11 |

**List of Tables**

|   |                                                                       |    |
|---|-----------------------------------------------------------------------|----|
| 1 | All functional area technical teams with team members listed. . . . . | ii |
| 2 | Avionics component values . . . . .                                   | 3  |
| 3 | Power demand of each subsystem . . . . .                              | 3  |
| 4 | Final cable parameters . . . . .                                      | 11 |
| 5 | Final cable parameters . . . . .                                      | 12 |
| 6 | Cable Insulator Properties . . . . .                                  | 15 |

# 1 Introduction

The aviation industry has set ambitious goals for reducing emissions of aircraft with Flightpath 2025 [1] being an example, so the need for alternative low or no emission solutions is high. Companies such as Airbus, Rolls Royce and many other large players in the aviation industry have turned to hydrogen to be a source of clean efficient fuel. The aircraft detailed in this report utilises fuel cell technology for electric propulsion. Fuel cell technology has the capacity to eliminate  $CO_2$  emissions however it is yet to be implemented on any commercial aircraft. The aim of this project is to act as a technology demonstrator for hydrogen fuel cell powered aircraft, with the goal of a solo circumnavigation making a strong case for the power source. This report will specifically focus on the design of the electrical systems on the aircraft. The electrical systems of a fuel cell powered aircraft differ from conventional combustion aircraft due to the high power needed to operate the systems. This report will investigate the need design and optimise the configuration and electrical components to allow for a viable design optimised for low mass.

To begin designing the electrical system the maximum continuous power drawn from each subsystem and hence the total maximum continuous power needed from the fuel cell is required. Methodology to size electrical components for each sub system is needed to allow for an accurate mass estimate. As well an investigation into redundancy and operation takes place improve reliability and usability of the system allowing it to successfully complete the five day mission. Finally key parameters such as fuel cell voltage is selected that complies with all constraints and results for the lowest mass. Then the placement of the electrical components on the aircraft can be optimised and the result giving the total electrical system mass and layout.

## 2 Power Sizing

For a hydrogen fuel cell powered aircraft there are five main branches of systems requiring power: propulsion, fuel cell compressor, cooling, environmental control systems (ECS) and Avionics. As mentioned in section 1 the first step is to determine the power requirements of each of these sub systems.

### 2.1 Propulsion

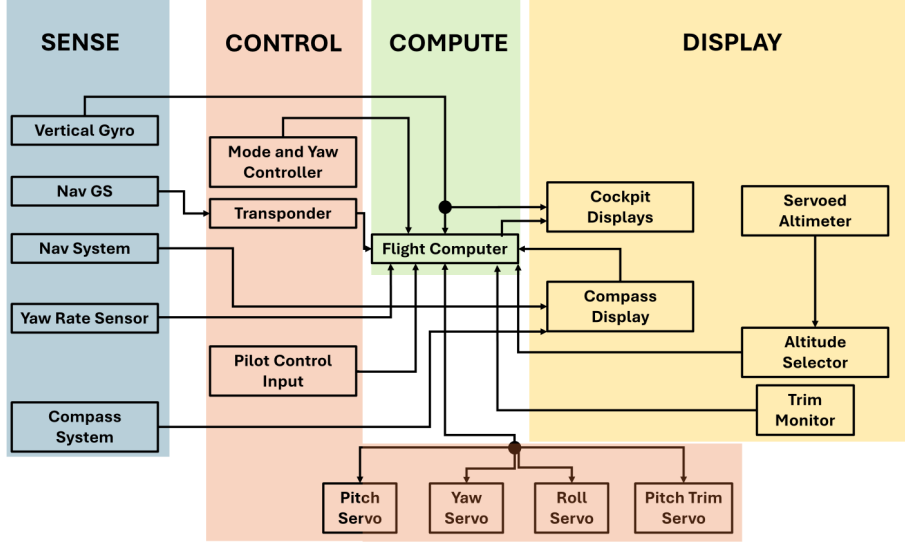
The propulsive power is determined by the flight regime and design of the aircraft. The power value was obtained with aid from the aerodynamics and mission planning teams through the development of a constraint diagram. The aircraft's power to weight ratio,  $(\frac{P}{W_0})$ , is  $5.76 \frac{W}{N}$  [2] for take off and a maximum takeoff weight of  $W_0 = 5860 \text{ kg}$  [3]. Maximum continuous propulsive power requirement is  $P_{prop} = 331 \text{ kW}$ . For two motors and a motor efficiency of 94 % [4] a motor power of  $P_{motor} = 176 \text{ kW}$  is required.

### 2.2 Avionics

The avionics system of the aircraft is very similar to that of the conventional lightweight solo pilot aircraft. Figure 1 shows the layout of the avionics systems on the aircraft, the design is taken from [5] which is deigned for a single pilot dual turboprop aircraft which is deemed a acceptable comparable to the circumnavigation aircraft.

Figure 1 highlights four main categories of avionics: sense, control, compute and display.

Sense describes the components that measure the state of the aircraft. The vertical gyro measures the aircraft's pitch and roll angles and with the yaw rate sensor allow for all three angles to be measured. The navigation units measure the aircraft's location. The compass system which is used to measure heading.



**Figure 1.** Layout of avionics systems on the aircraft

The mission is five days long and solo piloted therefore the use of an autopilot is needed. The previous sensors are used to allow both the pilot and autopilot to control the aircraft. The actuation system selected by [3] was a mechanical system with servos integrated on the cables to allow for autopilot control. The lightweight aircraft allowed for a mechanical system and removes the need for the same levels of redundancy of a fly by wire system.

The cockpit displays relay information to the pilot to allow them to control and monitor the aircraft. Engine information will be displayed through a custom Engine Instrument Display System (EIDS). The main difference for the avionics for a fuel cell powered aircraft is the interface between the pilot and fuel cell. The EIDS should display information about the electric motors and fuel cell stack. The electric motor parameters to be displayed are torque, RPM and temperature. For the fuel cell the temperature, fuel flow rate, voltage and current to ensure the stack is operating correctly. As well conventional displays will be integrated showing attitude and guidance, airspeed, vertical speed, heading, altitude and flight modes. Four cockpit displays were chosen as this is similar to conventional aircraft.

Table 2 tabulates all the avionics components showing individual masses and powers, this is then totaled and used in the power sizing process. The quantity of each component is selected based on its function. There are 3 of each sensor to allow for comparison of measurements and two of most other components for redundancy reasons. The result of the avionics power shows that it will play an insignificant role in the design of the electrical system as its power is much lower than the propulsive power, as well all of the components are rated at a low voltage of 28 VDC.

### 2.3 Environmental Control Systems

The environmental control systems maintains the pressure, temperature and other parameters important for pilot safety in the cockpit. The ECS maximum continuous power is 17.36 kW [4].

### 2.4 Thermal Management

One of the main issues with a fuel cell powered aircraft is the rejected heat during operation. The amount of heat is substantial and takes a dedicated heat exchanger system to resolve it. Due to this large design task the heat exchanger is detailed in [13]. The maximum continuous power to run the heat exchanger was estimated to be 1 kW.



**Table 2.** Avionics component values

| Part                        | Code          | Mass / kg    | Power / W     | Quantity |
|-----------------------------|---------------|--------------|---------------|----------|
| Vertical Gyro [5]           | KVG 350       | 2.72         | 88.00         | 3        |
| Nav Com [6]                 | TX56          | 2.00         | 10.00         | 2        |
| Yaw Rate Sensor [5]         | KRG 331       | 0.34         | 4.50          | 3        |
| Compass System [5]          | KCS 305       | 2.54         | 20.00         | 3        |
| Mode and Yaw Controller [5] | KC 290        | 0.22         | 20.00         | 2        |
| Transponder [7]             | GTX345        | 1.45         | 20.00         | 2        |
| Flight Computer [8]         | CDU-7000      | 4.94         | 70.00         | 1        |
| Cockpit Displays [9]        | CMA 6800      | 2.72         | 30.00         | 4        |
| Trim Monitor [5]            | KA 136        | 0.22         | 50.00         | 1        |
| Altitude Selector [5]       | KAS 297       | 0.54         | 10.00         | 1        |
| Servoed Altimeter [5]       | KEA 346       | 1.31         | 3.00          | 1        |
| Autopilot [10]              | AEROCRUZE 230 | 1.18         | 34.00         | 2        |
| Radio [11]                  | TY96          | 0.95         | 10.00         | 2        |
| Audio Panel [12]            | TMA44         | 0.60         | 10.00         | 1        |
| <b>Total</b>                | -             | <b>46.89</b> | <b>754.50</b> | -        |

## 2.5 Fuel Cell Compressor

The fuel cell reacts hydrogen and oxygen gas at certain pressures, therefore as altitude changes the ambient air pressure will not be sufficient. This necessitates the use of a compressor to increase the pressure of the air. There is no turbine being driven by hot exhaust gas to power the compressor, therefore the compressor is driven by an electric DC motor. The compressor power requirements are directly related to the fuel cell output power and the sizing methodology is detailed in [14]. The result is a fuel cell compressor maximum continuous power requirement of 46 kW [14].

## 2.6 Summary

**Table 3.** Power demand of each subsystem

| System               | Power / kW    | Proportion / % |
|----------------------|---------------|----------------|
| Propulsion           | 352.23 [2]    | 84.4           |
| Avionics             | 0.75          | 0.18           |
| ECS                  | 17.36 [4]     | 4.16           |
| Thermal Managment    | 1.00 [13]     | 0.26           |
| Fuel Cell Compressor | 46.00 [14]    | 11.0           |
| <b>Total</b>         | <b>417.34</b> | -              |

Table 3 summarises all the maximum continuous power requirements of each sub system. Propulsion is the largest power draw and therefore will contribute the most to the overall system mass. Therefore the design process will be heavily influenced by the propulsion system. Avionics and thermal management do not have a significant power draw and will not contribute a large amount to the overall system mass. Due to this the report will not detail the sizing and design of avionics and thermal management electrical systems.

## 3 Electrical Component Sizing

The required power load needed for each system is now quantified which allows for sizing of electrical components through out the aircraft. The electrical components used are similar to that of traditional

aircraft with: cables, busbars and circuit breakers. The fuel cell produces DC voltage and therefore the system will need DC/DC converters instead of the typical AC/DC converters used on traditional aircraft. As well as shown in section 2 the power ratings are much higher than traditional aircraft so proven aircraft components cannot be selected. Due to this an investigation into methods to size the components was carried out and detailed in this section.

### 3.1 Cables

The cables in the aircraft transfer power between systems and are composed of a conductor in the centre and an insulation sheath surrounding the conducting core. For a fuel cell powered aircraft the cable weights are considerably larger than in traditional combustion aircraft, this is due to the high power loads experienced for this type of aircraft. As well, due to the single fuselage design the cable lengths can be long increasing the cable weight more since systems can be placed far apart throughout the fuselage.

There are three main concerns when selecting voltage and current ratings for cables: cable mass, voltage safety and cable temperature. Cable mass is dictated by both current and voltage through conductor and insulator size. Voltage safety dictates insulator size and cable temperature is dependent on current and the material properties of the conductor.

To calculate the conductor area,  $A_c$ , the Equation 1 [15] is used and is a function of resistivity,  $\rho_{copper}$ , length of cable,  $L_{cable}$ , cable current,  $I_{cable}$  and the voltage drop,  $V_{drop}$ . From Equation 1, an increase in either cable current or cable length or a decrease in voltage drop will result in a larger conductor area and thus a heavier cable weight.

$$A_c = \frac{\rho_{copper} L_{cable} I_{cable}}{V_{drop}} \quad (1)$$

To calculate the insulation thickness,  $t_{insu}$ , of the cable Equation 2 [16] is used. Equation 2 [16] shows that the insulation thickness is a function of the conductor radius,  $r_c$ ,  $K$  a coefficient determined by the shape of the cavity, the cable voltage,  $V_{cable}$ , the maximum cavity thickness,  $t_{max}$  and the minimum voltage on the Paschen curve,  $\alpha$ . The value of  $K$  is determined through Equation 3 which is the expression for spherical voids. As well,  $K$  is a function of relative dielectric permittivity,  $\epsilon_r$ , which is a property of the insulation material.

$$t_{insu} = r_c \left( e^{\frac{KV_{cable}t_{max}}{\alpha r_c}} - 1 \right) \quad (2) \quad K = \frac{3\epsilon_r}{1 + 2\epsilon_r} \quad (3)$$

Using both Equation 1 and Equation 2 the cable is fully defined and along with the cable length and material properties the cable mass can be calculated.

To assess the viability of a cable Equation 4 is used. Equation 4 describes how the change in cable temperature,  $\Delta T_{cable}$  is determined by the cable current and voltage drop also expressed as power loss as well as the specific heat of copper,  $C_{p,copper}$ . This from the specific enthalpy equation and cable power equation. The result of this places constraints on the voltages, currents and insulator materials used throughout the system. These constraints are a large component as to why the electrical system mass is large.

$$\Delta T_{cable} = \frac{I_{cable} V_{drop}}{C_{p,copper}} \quad (4)$$

From Equation 1 and Equation 2 there are multiple parameters that are constant and selected. Firstly, the voltage drop is set and is the allowable amount of power loss through the cable, power loss is directly correlated with cable heat as shown in Equation 4. Due to this for certain sections of the aircraft, for example near the cryogenic tanks, setting a lower voltage loss and reducing heating is desirable.

Therefore the cable voltage drop was selected to be between 0.15 - 1% depending on the location of the cable. Dielectric permittivity is a property of the insulation material therefore will be dictated by the heat of the cables since certain insulating materials have different operating temperatures. A variety of powers are being managed by the system so a range of temperatures is experienced throughout, so multiple insulation materials are selected for each application. The selected insulation materials are, PTFE, PVDF, EPR with their properties summarised in Appendix A. Maximum cavity size was chosen from literature where a range of 3.5 - 50  $\mu m$  [16] was given, it was found that this value had little effect on the mass of the cables so the uncertainty on its value should not lead to a great disparity in results in an actual application, a value of 30  $\mu m$  was chosen as in the literature [16] this was used and allowed an ease of verification that the model used was correct. Finally the minimum voltage on the Paschen curve is a value that is dictated by the material surrounding your cable for this application this was taken to be air resulting in a value of 0.340 kV [16].

### 3.2 Busbars

A busbar is used to distribute current and power to different channels. Each bar has a voltage rating since all the components connected directly to a bus bar will have the same voltage as the busbar. Previously shown was a model of cable weight as a function of current and voltage therefore the selection of busbar voltages is critical in determining the weight of the electrical system, the selection of these voltages is detailed in section 5.

To size the busbar only the current,  $I_{bus}$  and number of connectors onto the bus,  $N_{bus}$ , are needed. The bus cross sectional area in square inches is found through Equation 5 [17] and it is standard to have the width be at least 3 times larger than the thickness [17]. Equation 6 [17] defines the bus length as a function of the cross sectional area, voltage drop, conductor resistivity,  $\rho$ , and bus current. All three dimensions of the bus are defined and the mass of the busbar can be calculated since the material used will be copper due to the materials favorable conductivity properties.

$$A_{bus} = 314E^{-6}(I_{bus})[1 + 0.05(N_{bus} - 1)] \quad (5) \quad L_{bus} = \frac{V_{drop}A_{bus}}{2\rho I_{bus}} \quad (6)$$

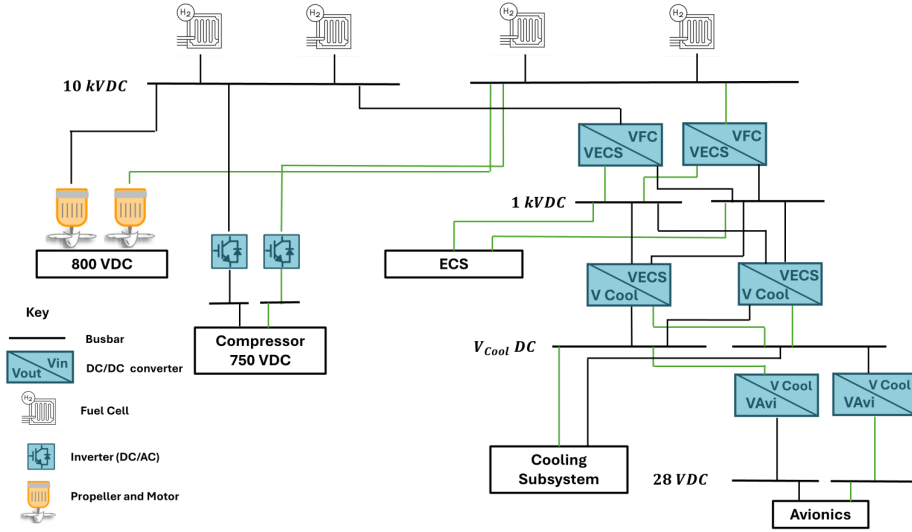
### 3.3 Voltage Converters

There are multiple systems all operating at different voltages so difficulty arises when the power source, in this case the fuel cell, produces power at a different voltage. To overcome this challenge power converters are implemented, which allow the voltage to be changed while keeping the power the same. For the electrical system of this aircraft DC/DC step down converters were used. The estimation of the mass of these components is important in the system design since at high power it can contribute a significant proportion to the overall system weight. The converters weight is estimated through power density values depending on the type of converter, DC/DC step down converters have a power density of 4 kW/kg [18]. This model only includes power as the only contributing factor towards converter weight, however in this application large voltages could be applied which can increase the weight of the converters through the need of greater insulation, resulting in an under estimate in converter mass using this methodology.

## 4 Redundancy and Operation

The goal of redundancy is to intentionally increase the number of components of a system to increase reliability. The electrical system on board the aircraft has to have multiple redundancies to ensure that in the event of a failure of certain components the aircraft is still operable and able to land safely. The operation of the electrical system includes how the system is interacted with by the pilot and other individuals working on the plane for example maintenance staff. This section will detail the decisions and considerations made for both of these areas and how this impacts the final design of the electrical system.

## 4.1 Redundancy



**Figure 2.** Electrical systems layout with redundancy considerations

The five systems detailed in section 2 are all critical as the plane would not be able to make a safe mission abortion in the event of losing complete power to one of these systems. As well the event of a failure should not cause a cascading effect where multiple systems are heavily effected in the event of a failure.

Figure 2 has no single point of failure with all components being doubled meeting the first criteria. As well the cross linking seen on the cascading branch on the right hand side of Figure 2. This design means the system will not experience a cascading failure along the branch therefore meeting the second criteria.

Current fuel cell technology needs to have a redundancy of 2.25 [19] so the fuel cell sub stacks in the system will meet this requirement. The main bus was split into two separate buses to avoid loss of power to all the systems in the event of a bus failure. These two buses provide power to all the systems in the aircraft and in the event of both of them failing the entire power to the aircraft is lost. Therefore an argument can be made to increase the number of main buses to further additional redundancy, this layout was not chosen because the bus bars are a low risk component with failure being rare during flight. Failure of buses usually occurs through installation error so this would be an error identified on the ground before takeoff. Secondly the additional weight cause by having to run an increasing number of high power cables is not worth the increase in redundancy for a component with a low level of failure during flight. In section 3 the bus bars were designed to be traditional copper plate bus bars, however these could be replaced with laminated bus bars. Laminated bus bars have multiple layers of copper separated by thin layers of dialectical material these bus bars have the capacity to increase reliability and prevent installation errors as well. At this stage of the design process traditional bus bars were much easier to implement and account for in terms of weight so laminated bus bars were not investigated further, however the weight difference between the two types is small so effect on overall weight estimate is not impacted greatly.

For the cascaded branch the event of failure to either the buses or transformers will cause a reduction in power supply to the systems. In this event the pilot should be alerted prompting an end to the mission. To alleviate the effects of this type of failure the system could throttle up the fuel cell. This has the effect of increasing power down the single operational branch which will then be able to distribute sufficient power to the systems. This will allow safe operation of the aircraft for enough time to land. However this approach will not be able to be deployed at segments of the mission corresponding to high power load as there is not enough extra power to provide. This is not an issue because this segment is the takeoff where it is easy for the pilot to abort in the event of a failure.

All traditional aircraft have an alternative power source in the event of power failure from the engines and system. For the fuel cell system there needs to be a back up as well because due to the single fuel tank design this results in a single point of failure for the system. The alternative power needs to be able to provide energy to the avionics and thermal management whilst the plane glides down to a diversion runway. The avionics is essential in allowing the pilot to monitor and fly the aircraft so will need to keep powered and the thermal management ensures that temperatures do not rise too much to cause further issues. ECS will not be powered due to the high power draw, so in the case of complete failure oxygen masks can be used by the pilot whilst the descent occurs. This is better than carrying a much larger battery over a long distance. The power can be stored in a battery onboard and is completely separate to the main electrical system. Using a lithium iron phosphate battery with an energy density of  $90 \frac{Wh}{kg}$  [20] and descent time of 80 minutes using the power needed for avionics and thermal management the estimated battery mass would be 30 kg. This is most likely an over estimate as the maximum continuous powers are used and a pessimistic energy density. However it demonstrates that the alternative energy system is not large.

A continuation into the redundancy of the electrical system would be an quantifiable analysis into the risk of failure during the five day mission. This would entail probability and reliability calculations for the components and therefore a change in system design in the event that this unveiled any issues.

## 4.2 Operation

The system is designed for an experimental aircraft planned to complete a single five day mission. Therefore the design of the electrical system is not heavily constrained by needs of fast turn around times on the ground and ease of checks and maintenance that a traditional commercial aircraft would have. However when placing electrical components and the systems consideration do have to be made to whether these components would be accessible for a technician to be able to replace or service the component. Since this is the first fuel cell aircraft electrical system there is a high likelihood of initial failure of the components and being able to service these would be useful for the aircraft's operation. This results in a consideration having to be made when positioning the components, avoiding areas deep into the after body since the volume of the fuselage decreases due to the up and down sweep taper. Also considering which components will need to be serviced, such as the fuel cell which may require sub stacks to be replaced in the event of failure.

Another aspect of the systems operation would be the connection to ground power on start up. Similar to traditional aircraft engines the fuel cell needs to power to start up before it can begin producing power. Therefore a ground power connection with the correct power output to start would need to be sized and integrated into the electrical system. The sizing of this is not within the scope of this report.

Previous circumnavigations have been piloted by two people, allowing for flying shifts and an additional layer of human redundancy to the system. In this mission since the pilot is solo the aircraft's condition will be monitored by a ground team. This allows for intervention on the mission if failures occur, allowing the ground to alert the pilot to any issues. The data transmitted to the ground would consist of pilot vital signs to ensure they are healthy and fit throughout the mission, as well as key aircraft data such as fuel cell conditions and aircraft position.

## 5 Electrical System Design

The overall systems, components and general layout of the electrical system for the aircraft have been described, quantified and modelled, now the design of the components and their subsequent positioning on the plane can be calculated and optimised. The goal of this is to create an optimal weight solution for the electrical system. To do this the first step is to define the fuel cell voltage and then place the main bus optimally along with the other components. The placement of these components however will impact other sub teams involved in the design of the aircraft so consideration of the effects of

placement on those teams is important. Mainly this is the concerns of stability handled by the control team. The placement of the systems was dictated strongly by the control team since the gains in electrical system mass did not outweigh the effects on stability. The electrical design was given the freedom to place the smaller electrical components such as cables, buses and converters optimally. This section will detail the rationale behind key design decisions for the electrical system.

## 5.1 Component Selection

In Figure 2 there are components that will be selected from other sub teams influencing the design of the electrical system. The motors were sized from the propulsive power as well as the torques and RPM needed to operate the propeller. This is detailed in [4], and the motor was designed based on the HPDM-250 a DC motor with a rated voltage of 800V [21]. The fuel cell compressor will be powered by a DC motor that will be sized based off of maximum fuel cell power output during the mission which requires a certain flow rate of air at a certain pressure. The details of the compressor are detailed in [14] with the compressor rated at 750 V. The ECS requires its own compressors these were sized in however specific compressors were not selected so a ECS bus voltage of 750 V was selected to match the fuel cell compressor however with more detailed analysis of the system a more appropriate voltage will be selected. The avionics bus voltage was set to 28 VDC which is a standard voltage level for avionics in aircraft, the investigation into the connections of each individual component and the required redundancy was not explored in this report due to the limited scope, as well due to the low power the weight contribution from the electronics for the avionics will be very insubstantial comparatively. In the same spirit the cooling system electronics were not accounted for due to the low power and lack of detail in electrical components at this stage of the design, again this system will have a small contribution to the mass.

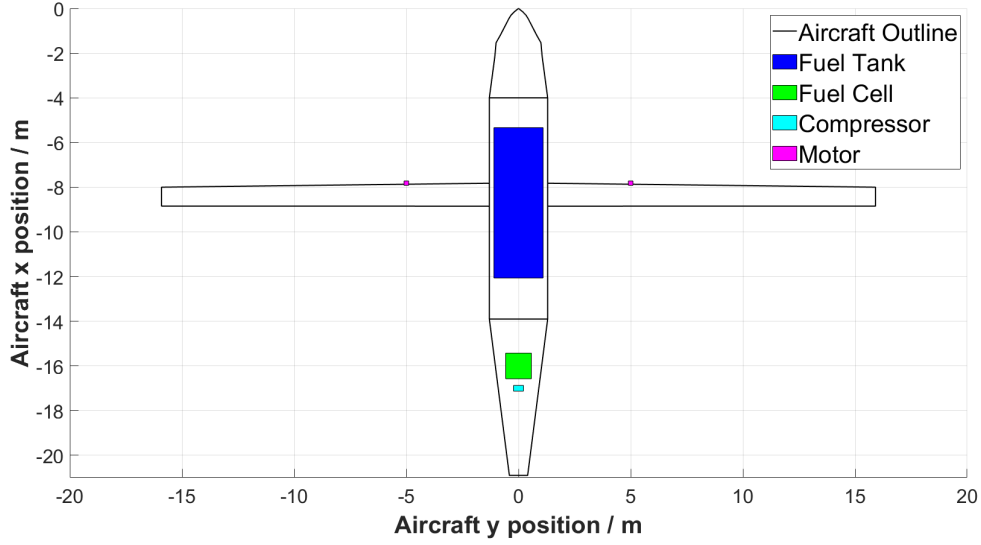
## 5.2 Fuel Cell Voltage

As described in subsection 3.1 the voltage and current levels through the cables decide the mass, with higher voltage and lower current leading to lighter cables. The weight of electrical systems is heavily dependent on the power load applied to the components, this means the high power electronics will contribute the most to the weight. Figure 2 the two main buses provide power to the whole system and therefore transfer the most power throughout the system than any other component. The main bus voltage will be the same as the fuel cell voltage so an appropriate selection of fuel cell voltage will result in an optimal mass of the system.

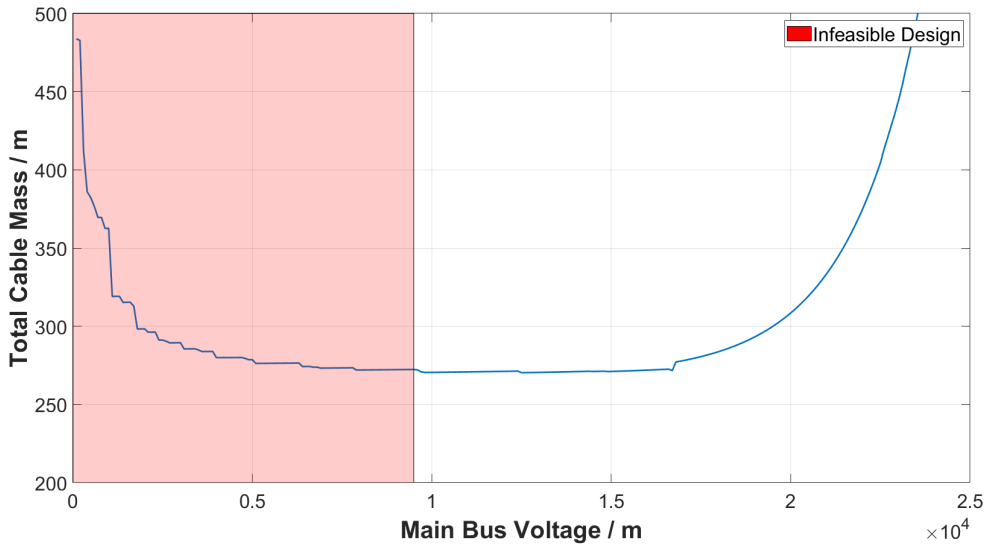
As shown in section 2 the propulsive systems required the most power. As described previously the HPDM-250 motors will be used they are DC [21] avoiding the requirement of a DC/AC converter and will be rated at 800 V. Therefore, any fuel cell voltage rating larger than the motor voltage rating will require a high power DC/DC step down converter. Therefore it was undesirable to increase the fuel cell voltage above 800 VDC and include the heavy converters. However, the cables would have very high currents leading to cable temperatures exceeding the limits of the insulation material. So the use of high power DC/DC converters had to be used for the propulsive branch due to the temperatures.

To begin to decide the fuel cell voltage the positions of components is needed to allow for cable lengths to be calculated. Figure 3 shows the position of the fuel cell, compressor, fuel tank and motors. These positions were decided through stability analysis conducted by the control team and the constraints placed on the positions by the systems team and detailed in a weight and balance report [3]. The converters were placed as close to their respective components to allow for the high voltage light weight wires to be as long as possible and the main bus placed centrally to service all systems. With these positions the investigation into fuel cell and system weight was conducted, Figure 4 shows the results.

Figure 4 shows the electrical system mass as fuel cell voltage varies, with the red region being where the currents are too high causing the cable temperatures to pass the limit of the materials. The general trend is what is expected, low voltages resulting in large conductor areas and a large mass, and then high voltages leading to thick insulation and heavier wires. There is a plateau over a large range of



**Figure 3.** Systems layout on the aircraft



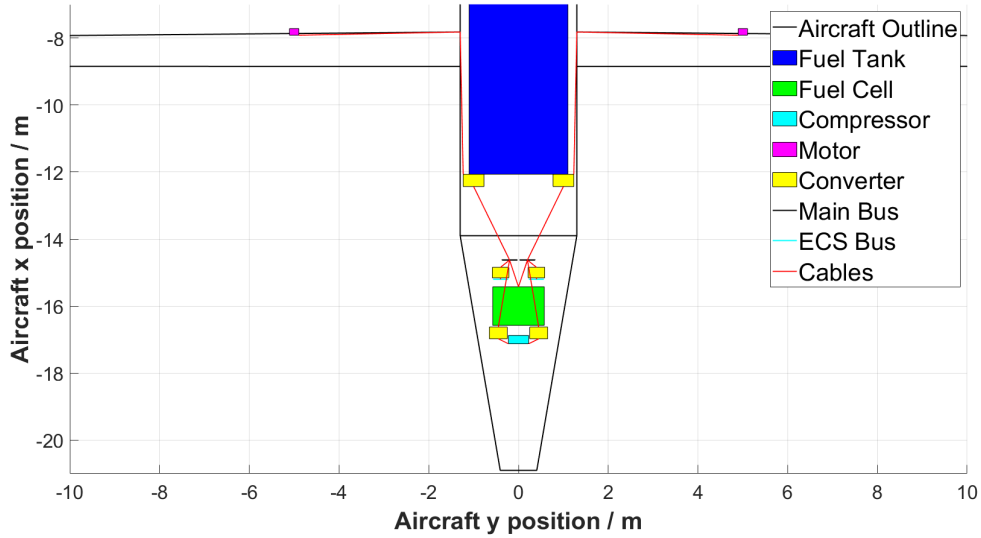
**Figure 4.** Electrical system mass as fuel cell voltage varies

voltages where the variation in mass is a few kilograms at most, this gives flexibility with the design choice as moving away from the most optimal voltage will not result in large mass gains. The optimal fuel cell voltage from Figure 4 is 14 kV, however the selected fuel cell voltage was 10 kV. The lower voltage was selected because the fuel cell stack is split into sub stacks arranged in parallel to achieve the voltage and current desired. At high voltages less sub stacks are needed since more fuel cells in parallel results in a higher voltage, however less sub stacks introduces risks around redundancy as a failure in a fuel cell could effect a much larger proportion of the power supply. As mentioned in section 4. A minimum of 2.25 sub stacks was needed for acceptable redundancy in the design and a fuel cell voltage of 14 kV was not able to provide that. Therefore, 10 kV was chosen as it met the thermal demands of the system and the fuel cell stack redundancy without too much additional mass. Also as mentioned in subsection 3.3 the converter model used neglected the voltage difference across the converter, so it is possible that the 14kV system would not be optimal and a selection of a lower voltage near the thermal limit is acceptable.

### 5.3 Electrical Component Position Optimisation

With the fuel cell voltage defined and the system electrical components selected the final part of the electrical system design at this stage is the placement of buses and converters. The main components to be placed are, the main buses, motor power converters, compressor power converters, ECS converters and buses.

The optimisation process works by selecting regions of the aircraft that each component can be placed and then calculating the mass of the whole system for each possible arrangement, finally outputting the arrangement that resulted in the lowest overall system mass. Referring back to Figure 3, the region for the main bus to be placed was between the fuel cell and fuel tank, since placing electrical components inside the integrated fuel tank is undesirable due to limited space, heating and difficulty in access for maintenance crews. The same region was applied to the motor power converter as well. The other components were placed manually since their contribution to overall system mass was low and this allowed for reduced complexity of the process. The results of this process are shown in Figure 5 where the components are to scale, for the power converters, similar power rated DC/DC step down converters [22] dimensions were selected.

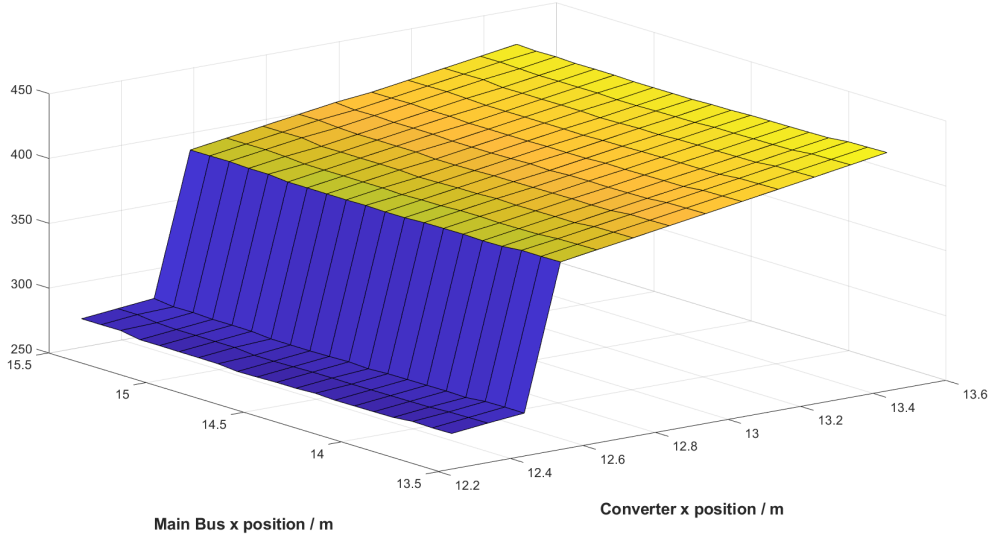


**Figure 5.** Optimised electrical component locations

The optimisation resulted in a total electrical system mass of 423 kg. The motor converters were placed as close to the fuel tanks as possible as the cable running from the converter to the motor contributed the most to the mass. Therefore reducing that distance by moving the converter more towards the wings, however due to the size of the converters and the structural implications of placing them in the wing was not feasible. The cables to the motor run along the leading edge due to the closed wing box design so the available space is through the D-section of the wing [23]. The main bus was placed centrally this allowed a balance in distance between all systems.

Figure 6 shows the results of the optimisation with the variation in electrical system cable mass with the position of the main bus and motor power converter. The result shows how sensitive the design is since a small movement of the motor power converter away from the fuel tank will result in a large increase in cable mass. This demonstrates the desire from an electrical systems perspective to create space to move that converter closer to the wings in future iterations, however this is limited by structural and mainly control concerns of the aircraft.





**Figure 6.** 3D plot showing system mass with varying main bus and motor power converter position

## 5.4 Summary

Table 4 shows the properties of the cables used through out the system. The operating conductor temperatures at maximum and average load are shown as well as the cable mass and insulator material used. The conductor temperatures are all within the continuous operating temperature of the respective insulator these values are shown in Appendix A. Additionally this maximum power condition is only for take off which is a very short part of the flight, so cable temperature is deemed acceptable.

**Table 4.** Final cable parameters

| Cable                | Mass / kg  | $T_{Max}$ / $^{\circ}C$ | $T_{Average}$ / $^{\circ}C$ | Insulator Material |
|----------------------|------------|-------------------------|-----------------------------|--------------------|
| Fuel Cell            | 1.15       | 149                     | 93                          | PVDF               |
| Motor Converter      | 5.56       | 246                     | 133                         | PTFE               |
| Motor                | 262        | 254                     | 137                         | PTFE               |
| Compressor Converter | 1.31       | 65                      | 53                          | EPR                |
| Compressor           | 0.49       | 105                     | 84                          | EPR                |
| ECS converter        | 0.12       | 42                      | 31                          | EPR                |
| ECS bus              | 0.11       | 218                     | 119                         | PTFE               |
| <b>Total</b>         | <b>271</b> | -                       | -                           | -                  |

Due to the fuel tank the motor power converters could not be placed closer to the motors. This resulted in very high current wires travelling over the long distance. To counteract this the design split the cable into 36 smaller cables to reduce the heating substantially, however this still causes concerns at output of the converter and the input of the motor where the 36 cables have to join together to deliver the required current. Due to the short time this occurs and the localised heating at the connection points the use of heat sinks could be used. Especially for the connection at the motor, the electric motor itself will also need to be cooled through air and heat sinks in the nacelle. So the wire cooling can be integrated with the electric motor cooling within the nacelle. Likewise the converters will have rejected heat that will need to be dissipated and the cable heat can be integrated and rejected in the converters system again potentially using a heat sink. However this needs to be investigated further to check if this is viable. If not then the implementation of superconductors could be needed, but this causes difficulty due to the operating conditions of current superconductor technologies. Alternatively the use of distributed power could be investigated as this would reduce the power requirement per

motor alleviating the loads on the cables.

Table 2 shows the mass and proportion of each electrical component in the system. The results align with the expectation of the propulsion system being the largest contributor to system weight and subsequently the cables being the largest component contributor. The total electronic system mass of 423 kg was greater than the estimate at the beginning of the iteration. This is because without knowing the placement of systems and a more detailed breakdown of system power draw it is difficult to estimate the mass, moreover this type of aircraft is the first of its kind therefore there are no benchmarks to measure the aircraft against.

**Table 5.** Final cable parameters

| Component    | Mass / kg  | Proportion / % |
|--------------|------------|----------------|
| Cables       | 271        | 64             |
| Converters   | 152        | 35.9           |
| Buses        | 0.04       | 0.1            |
| <b>Total</b> | <b>423</b> | -              |

## 6 Conclusion

For the mission and configuration of this aircraft the report outlines a viable initial design of the electrical system. The systems overall power demand, layout, weight estimation and placement on the aircraft is initially designed.

Power demand was collated from many members of the sub team and this report detailed the design of the avionics system and its associated power draw. The propulsion sub system has the largest power draw as expected, with 81.6 % of the total power with the total fuel cell power being 417 kW.

The electrical system design was influenced by redundancy and operation. This required the duplication of many components resulting in increase in mass. The mass of a back up battery power source was estimated to be 30 kg. The operation of the electrical system was also outlined, including maintenance accessibility, ground power integration and use of a ground station to monitor the aircraft during the mission. The constraints from operation were weak due to the experimental nature of the aircraft.

The electrical system was designed around a fuel cell voltage of 10 kV, this resulted in the lowest mass whilst still complying with thermal constraints. The system was laid out on the aircraft with the components placed in optimal locations. The optimisation approach was to investigate the placement of the propulsion components as this had the largest effect on mass. This resulted in a total electrical systems mass of 423 kg with the cables being the largest proportion of the mass at 64 %.

The goal of this aircraft is to be a technology demonstrator for hydrogen powered aircraft. The mission and experimental nature of the aircraft allowed certain freedoms in the design which will not be afforded for integration into a commercial airliner. The space available for components was much larger due to no passengers and the power requirement much lower than that for a large aircraft. With a much larger power requirement it would not be possible to design cables to operate at acceptable temperatures without the use of super conductors or distributed propulsion.

Additional work into the effects of cable and converter temperature and the cooling methods needed such as heat sinks need to be quantified if this possible. The current configuration allows for operation within the insulator material limits for cables between connections. As well further redundancy analysis into potential failure modes with more quantitative analysis would be required. The design of the avionics and thermal management electrical systems would also be needed.

## References

- [1] European Commission. Flightpath2050:Europe’s vision for aviation; maintaining global leadership and serving society’s needs; 2011. Available from: <https://doi.org/10.2777/50266>.
- [2] Ji B. Constraint diagram, airfoil selection and wing design. Hydrogen Solo Circumnavigation. 2024.
- [3] Collins N. Weight Estimation and Subsystems. Hydrogen Solo Circumnavigation. 2024.
- [4] Ali MMA. Environmental Control System, Emergency Systems and Propeller Design. Hydrogen Solo Circumnavigation. 2024.
- [5] Bendix. Pilot’s Guide KFC 250; 1990. Available from: [https://www.bendixking.com/content/dam/bendixking/en/documents/downloads/006-08305-0001\\_1-krq-331-pilot-s-guide.pdf](https://www.bendixking.com/content/dam/bendixking/en/documents/downloads/006-08305-0001_1-krq-331-pilot-s-guide.pdf).
- [6] Trig Avionics. TX56/56A and TX57/57A Installation Manual; 2022.
- [7] Garmin. GTX™ 345; 2022. Available from: <https://www.garmin.com/en-US/p/140949#specs>.
- [8] Rockwell Collins Inc. CDU 7000; 2008. Available from: [https://prd-sc102-cdn.rtx.com/-/media/ca/product-assets/files/public/products/product-brochures/communication-and-networks/comm\\_audio-mgmt-controls/cdu-7000-data-sheet.pdf?rev=5bf355fd17ad401db54b256d46c286fe&hash=B2CC63E34F497660A64A7A5F953C3F4B](https://prd-sc102-cdn.rtx.com/-/media/ca/product-assets/files/public/products/product-brochures/communication-and-networks/comm_audio-mgmt-controls/cdu-7000-data-sheet.pdf?rev=5bf355fd17ad401db54b256d46c286fe&hash=B2CC63E34F497660A64A7A5F953C3F4B).
- [9] Honeywell International Inc. CMA 6800 COCKPIT DISPLAY; 2021. Available from: <https://aerospace.honeywell.com/us/en/products-and-services/product/hardware-and-systems/cockpit-systems-and-displays/cma-6800-cockpit-display>.
- [10] BendixKing. AeroCruze 230 Advanced Touchscreen, 2 or 3 Axis Autopilot; 2020. Available from: <https://aerospace.honeywell.com/content/dam/bendixking/en/documents/downloads/n61-2106-000-000-aerocruze-230-brochure.pdf>.
- [11] Trig Avionics. TY96/96A and TY97/97A VHF Radio Installation Manual; 2019. Available from: <https://trig-avionics.com/library/01238-00-AK%20TY96%20and%20TY96A%20Installation%20Manual.pdf>.
- [12] Trig Avionics. TMA44 Audio Panel Installation Manual; 2018. Available from: <https://trig-avionics.com/library/01849-00-AD%20TMA44%20Installation%20Manual%20Web.pdf>.
- [13] Jalal-Aldeen H. Thermal Management Systems. Hydrogen Solo Circumnavigation. 2024.
- [14] Coyde D. Fuel Cell Sizing and Design. Hydrogen Solo Circumnavigation. 2024.
- [15] Britannica. "resistivity"; 2024. Available from: <https://www.britannica.com/science/resistivity>.
- [16] Cheng FC. Insulation Thickness Determination of Polymetric Power Cables. Department of Electrical Engineering, Hong Kong Polytechnic. 1994.
- [17] Mersen. Design Guide Formulas;. Available from: <https://ep-us.mersen.com/products/engineering/design-guide-formulas>.
- [18] Brombach J. Comparison of Different Electrical HVDC Architectures for Aircraft Application. Institute of Electrical Power Systems. 2012.
- [19] Kösters TL, Gao X. Safety certification and time constant determined extended operations for fuel cell powered aircrafts. CEAS Aeronaut. 2021.
- [20] FluxPower. What is the Energy Density of a Lithium-Ion Battery?;. Available from: <https://www.fluxpower.com/blog/what-is-the-energy-density-of-a-lithium-ion-battery>.

- [21] H3X. HPDM-250 Datasheet;. Available from: <https://h3x.docsend.com/view/swr728yrvveakgfg>.
- [22] Zekalabs. RedPrime DC-DC Converter;. Available from: <https://zekalabs.com/products/non-isolated-high-power-converters/dc-dc-converter-200kw-1200v/>.
- [23] Tolley M. Simplified Analytical Wing Model, Material Selection and Manufacturing Feasibility. Hydrogen Solo Circumnavigation. 2024.
- [24] Microwaves101. PTFE;. Available from: <https://www.microwaves101.com/encyclopedias/ptfe>.
- [25] Junkosha. Allowable Current Calculation Formula;. Available from: <https://www.junkosha.com/en/innovation/technical/wire03>.
- [26] DesignerData. PVDF;. Available from: <https://designerdata.nl/materials/plastics/thermo-plastics/polyvinylidene-fluoride?cookie=YES>.
- [27] Tsyganov A, Vikulova MA. Significantly Enhanced Balance of Dielectric Properties of Polyvinylidene Difluoride Three-Phase Composites by Silver Deposited on K<sub>2</sub>Ni<sub>0.93</sub>Ti<sub>7.07</sub>O<sub>16</sub> Hollandite Nanoparticles. MDPI. 2024.
- [28] MatWeb. Ethylene Propylene Rubber (EPM, EPDM);. Available from: <https://www.matweb.com/search/datasheet.aspx?matguid=f8e3355cc2c541fbb0174960466819c0>.
- [29] Cao L, Grzybowski S. Life-time characteristics of EPR cable insulation under electrical and thermal stresses. IEEE International Conference on Solid Dielectrics (ICSD). 2013.

# Appendix

## A Cable Insulator Material Properties

**Table 6.** Cable Insulator Properties

| Material | Density / $\frac{kg}{m^3}$ | Dielectric Permeability ( $\epsilon_r$ ) | $T_{max,continuous}$ / $^{\circ}C$ |
|----------|----------------------------|------------------------------------------|------------------------------------|
| PTFE     | 2200 [24]                  | 2.02 [24]                                | 260 [25]                           |
| PVDF     | 1770 [26]                  | 10 [27]                                  | 150 [25]                           |
| EPR      | 860 [28]                   | 3.5 [16]                                 | 105 [29]                           |